$$\text{Median} = l + \left(\frac{\frac{N}{2} - m}{f}\right) \times c$$

$$\frac{N}{2} = \frac{140}{2} = 70$$

Here $l = 49.5,\ f = 30,\ m = 65,\ c = 10$

$$\text{Median} = 49.5 + \left(\frac{70 - 65}{30}\right) \times 10$$

$$= 49.5 + 1.67$$

$$= 51.17$$

Graphical method for Location of median

Median can be located with the help of the cumulative frequency curve or 'ogive'. The procedure for locating median in a grouped data is as follows:

Step 1 : The class intervals, are represented on the horizontal axis (x-axis)

Step 2 : The cumulative frequency corresponding to different classes is calculated. These cumulative frequencies are plotted on the vertical axis (y-axis) against the upper limit of the respective class interval

Step 3 : The curve obtained by joining the points by means of freehand is called the *'less than ogive'*.

Step 4 : A horizontal straight line is drawn from the value $\frac{N}{2}$ or $\frac{N+1}{2}$ on the y-axis parallel to x- axis to meet the ogive. (depending on N is odd or even)

Step 5 : From the point of intersection, draw a line, perpendicular to the horizontal axis which meet the x axis at m say.

Step 6 : The value m at x axis gives the value of the median.

Remarks:

(i) Similarly 'more than' ogives, can be drawn by plotting more than cumulative frequencies against lower limit of the class. A horizontal straight line is drawn from the value $\frac{N}{2}$ or $\frac{N+1}{2}$ on the y-axis parallel to x-axis to meet the ogive. A line is drawn perpendicular to x-axis meets the point at m, say, the X coordinate of m gives the value of the median.

(depending on N is odd or even)

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